NAMANGA SECONDARY SCHOOL

FORM FOUR — INFORMATION AND COMPUTER STUDIES

DATABASE PRACTICAL EXAMINATION

MARKING SCHEME AND STEP-BY-STEP GUIDE

Total Marks: 100

This guide gives the expected answers and simple Microsoft Access steps for every question in the examination.

> Marking rule: Give the mark when the work is correct, even if the candidate uses another simple correct Access method.

Section	Topic	Marks
A	Database creation	5
B	Table design and data entry	20
C	Relationship	8
D	Validation and Lookup Wizard	10
E	Queries	30
F	Sorting and filtering	5
G	Forms	6
H	Reports	6
I	Testing	6
J	Database check	2
K	Final submission	2
TOTAL		100

QUESTION A — DATABASE CREATION (5 MARKS)

A(a) Create and save the database — 2 marks

Expected answer/result: Database name: NAMANGA_STUDENT_DATABASE.

1. Open Microsoft Access.
2. Choose Blank database.
3. Enter NAMANGA_STUDENT_DATABASE.
4. Click Create.

Marking: 1 mark for creating the database; 1 mark for the correct name.

A(b) Create the three tables — 3 marks

Expected answer/result: Create STUDENT, CLASS and ACTIVITY.

1. Go to Create → Table Design.
2. Create the fields for STUDENT and save it as STUDENT.
3. Create the fields for CLASS and save it as CLASS.
4. Create the fields for ACTIVITY and save it as ACTIVITY.

Marking: 1 mark for each correct table.

QUESTION B — TABLE DESIGN AND DATA ENTRY (20 MARKS)

Use Design View. The field names, suitable data types, primary keys and records must be correct.

1. STUDENT table

Field	Data type	Key / note
StudentID	Short Text	Primary Key
FirstName	Short Text	
LastName	Short Text	
Gender	Short Text	Lookup: Male, Female
DOB	Date/Time	
ClassCode	Short Text	Foreign Key
Phone	Short Text	10 digits
Mark	Number	0–100

StudentID	FirstName	LastName	Gender	DOB	ClassCode	Phone	Mark
ST001	Amani	Mollel	Male	12/03/2010	F2A	0712345678	78
ST002	Neema	Joseph	Female	27/07/2011	F2A	0756789012	84
ST003	Musa	Kweka	Male	19/01/2009	F3A	0765432109	67
ST004	Halima	Said	Female	05/11/2010	F2B	0687654321	72
ST005	Baraka	Mushi	Male	22/09/2008	F3A	0711223344	91
ST006	Maria	John	Female	14/02/2012	F1A	0744556677	58
ST007	Kelvin	Mrema	Male	30/06/2011	F2B	0788990011	63
ST008	Rehema	Nnko	Female	17/08/2009	F3B	0622334455	88
ST009	Omari	Juma	Male	02/05/2010	F2A	0699887766	49
ST010	Asha	Mwakalinga	Female	25/12/2008	F3B	0714556677	76

2. CLASS table

Field	Data type	Key / note
ClassCode	Short Text	Primary Key
ClassName	Short Text	
Stream	Short Text	
RoomNo	Short Text	
ClassTeacher	Short Text	

ClassCode	ClassName	Stream	RoomNo	ClassTeacher
F1A	Form I	A	R01	Mr Kato
F2A	Form II	A	R02	Ms Neema
F2B	Form II	B	R03	Mr Baraka
F3A	Form III	A	R04	Ms Asha
F3B	Form III	B	R05	Mr Musa

3. ACTIVITY table

Field	Data type	Key / note
ActivityID	Short Text	Primary Key
ActivityName	Short Text	
Category	Short Text	
ActivityDate	Date/Time	
Participants	Number	
Fee	Currency/Number	

ActivityID	ActivityName	Category	ActivityDate	Participants	Fee
AC01	Cyber Safety Talk	ICT	20/01/2026	80	0
AC02	Football Match	Sports	24/01/2026	96	500
AC03	Database Club	Academic	28/01/2026	24	0
AC04	Science Fair	Academic	13/02/2026	55	1000
AC05	Girls in ICT Day	ICT	28/02/2026	45	0
AC06	Netball Match	Sports	07/03/2026	60	300
AC07	ICT Workshop	ICT	21/03/2026	38	2000
AC08	Programming Challenge	ICT	28/03/2026	27	500

Simple steps for all three tables:

1. Go to Create → Table Design.
2. Type the field names and choose the data types.
3. Select the key field and click Primary Key.
4. Save with the exact table name.
5. Open Datasheet View and enter the given records.

Marking: award marks for correct fields/data types, correct primary keys and correct data entry. Section B = 20 marks.

QUESTION C — RELATIONSHIP (8 MARKS)

Expected relationship: CLASS.ClassCode → STUDENT.ClassCode. One class can have many students.

1. Close tables if needed.
2. Go to Database Tools → Relationships.
3. Add CLASS and STUDENT.
4. Drag ClassCode from CLASS to ClassCode in STUDENT.
5. Tick Enforce Referential Integrity.
6. Click Create.
7. Check the relationship line.
8. Do not create a relationship for ACTIVITY.

Item	Expected work	Marks
C(a)	CLASS.ClassCode linked to STUDENT.ClassCode	3
C(b)	ClassCode is primary key in CLASS and foreign key in STUDENT	2
C(c)	Enforce Referential Integrity selected	2
C(d)	No relationship for ACTIVITY	1
TOTAL		8

Quick check: entering ClassCode F9A in STUDENT should be rejected because F9A is not in CLASS.

QUESTION D — VALIDATION AND LOOKUP (10 MARKS)

D(a) Gender Lookup Wizard — 2 marks

1. Open STUDENT in Design View.
2. Select Gender.
3. Choose Lookup Wizard.
4. Choose to type the values yourself.
5. Enter Male and Female.
6. Set the field to allow only the listed values.

   Expected answer/result: Only Male and Female are allowed.

D(b) ClassCode Lookup Wizard — 2 marks

1. Select ClassCode in STUDENT.
2. Choose Lookup Wizard.
3. Choose values from another table/query.
4. Choose CLASS.
5. Select ClassCode.
6. Finish and save.

   Expected answer/result: The drop-down should show F1A, F2A, F2B, F3A and F3B.

D(c) Mark validation — 2 marks

1. Open STUDENT in Design View.
2. Select Mark.

3. Enter Between 0 And 100 in Validation Rule.
4. Save.

   Expected answer/result: Validation Rule: Between 0 And 100.

D(d) Phone — 1 mark

Expected answer/result: Set Phone to Short Text. Validation Rule: Like "0#########".

D(e) Participants — 1 mark

1. Open ACTIVITY in Design View.
2. Select Participants.
3. Enter >=1 in Validation Rule.
4. Save.

   Expected answer/result: Validation Rule: >=1.

D(f) Fee — 1 mark

1. Select Fee.
2. Enter >=0 in Validation Rule.
3. Save.

   Expected answer/result: Validation Rule: >=0.

D(g) ActivityDate — 1 mark

1. Select ActivityDate.
2. Enter Between #01/01/2026# And #31/12/2026#.
3. Save.

   Expected answer/result: Dates outside 2026 should be rejected.

QUESTION E — QUERIES (30 MARKS)

General method: Create → Query Design → add table(s) → add fields → enter Criteria → Run → save.

E1. Q_AllStudents — 1 mark

1. Add STUDENT.
2. Add StudentID, FirstName, LastName and Mark.
3. Click Run.
4. Save as Q_AllStudents.

Expected answer/result: Criteria/formula: No criteria.
Result: All students.

E2. Q_NameStartsM — 1 mark

1. Add STUDENT.
2. Add the needed fields.
3. Under FirstName type Like "M*".
4. Run and save as Q_NameStartsM.

Expected answer/result: Criteria/formula: Like "M*"
Result: FirstName starts with M; expected: Musa.

E3. Q_LastNameAR — 1 mark

1. Add STUDENT.
2. Add LastName.
3. Type Like "*ar*" under LastName.
4. Run and save as Q_LastNameAR.

Expected answer/result: Criteria/formula: Like "*ar*"
Result: LastName contains ar.

E4. Q_ClassIN — 1 mark

1. Add STUDENT.
2. Add the needed fields.
3. Under ClassCode type In ("F2A","F2B","F3A").
4. Run and save as Q_ClassIN.

Expected answer/result: Criteria/formula: In ("F2A","F2B","F3A")
Result: Students in F2A, F2B or F3A.

E5. Q_Mark75Plus — 1 mark

1. Add STUDENT.
2. Add the needed fields.
3. Under Mark type >=75.
4. Run and save as Q_Mark75Plus.

Expected answer/result: Criteria/formula: >=75
Result: Mark 75 or above.

E6. Q_Mark60_80 — 1 mark

1. Add STUDENT.
2. Add the needed fields.

3. Under Mark type Between 60 And 80.
4. Run and save as Q_Mark60_80.

   Expected answer/result: Criteria/formula: Between 60 And 80
   Result: Mark from 60 to 80.

E7. Q_Female70Plus — 2 marks

1. Add STUDENT.
2. Add Gender and Mark.
3. Under Gender type "Female".
4. On the same row put >=70 under Mark.
5. Run and save as Q_Female70Plus.

   Expected answer/result: Criteria/formula: Gender="Female" AND Mark>=70
   Result: Female students with Mark 70 or above.

E8. Q_MaleOR90 — 2 marks

1. Add STUDENT.
2. Add Gender and Mark.
3. Put "Male" on the first criteria row.
4. Put >=90 on the OR row under Mark.
5. Run and save as Q_MaleOR90.

   Expected answer/result: Criteria/formula: Gender="Male" OR Mark>=90
   Result: Male students OR any student with Mark 90 or above.

E9. Q_DOB_2009_2010 — 2 marks

1. Add STUDENT.
2. Add DOB and needed fields.
3. Under DOB type Between #01/01/2009# And #31/12/2010#.
4. Run and save as Q_DOB_2009_2010.

   Expected answer/result: Criteria/formula: Between #01/01/2009# And #31/12/2010#
   Result: Students born in 2009 or 2010.

E10. Q_Older70Plus — 2 marks

1. Add STUDENT.
2. Add DOB and Mark.
3. Under DOB type <#01/01/2010#.
4. On the same row put >=70 under Mark.
5. Run and save as Q_Older70Plus.

   Expected answer/result: Criteria/formula: DOB < #01/01/2010# AND Mark >=70
   Result: Born before 01/01/2010 and Mark 70 or above.

E11. Q_Age16Plus — 3 marks

1. Add STUDENT.
2. Add StudentID, FirstName, LastName and DOB.
3. Create a calculated field called Age using the formula below.
4. Put >=16 in the Age Criteria.
5. Run and save as Q_Age16Plus.

Expected answer/result: Criteria/formula:
DateDiff("yyyy",[DOB],Date())-IIf(Format([DOB],"mmdd")>Format(Date(),"mmdd"),1,0) ; Age >=16
Result: Shows students aged 16 or more.

E12. Q_FormIII — 2 marks

1. Add STUDENT and CLASS.
2. Keep the relationship line.
3. Add FirstName, LastName, ClassName, Stream and Mark.
4. Under ClassName type "Form III".
5. Run and save as Q_FormIII.

Expected answer/result: Criteria/formula: ClassName="Form III"
Result: Form III students with class details and marks.

E13. Q_Performance — 3 marks

1. Add STUDENT.
2. Add StudentID, FirstName, LastName and Mark.
3. Create PerformanceBand using the formula below.
4. Run and save as Q_Performance.

Expected answer/result: Criteria/formula:
IIf([Mark]>=80,"Excellent",IIf([Mark]>=60,"Good","Needs Improvement"))
Result: 80+ Excellent; 60–79 Good; below 60 Needs Improvement.

E14. Q_MinimumMark — 2 marks

1. Add STUDENT.
2. Add the needed fields including Mark.
3. Under Mark type [Enter minimum Mark:].
4. Run and test with 70.
5. Save as Q_MinimumMark.

Expected answer/result: Criteria/formula: [Enter minimum Mark:]
Result: Asks the user for a minimum mark.

E15. Q_ClassGender — 2 marks

1. Add STUDENT and CLASS.
2. Choose Crosstab Query.
3. Use ClassName as Row Heading.
4. Use Gender as Column Heading.
5. Use StudentID as Value and Count it.
6. Run and save as Q_ClassGender.

Expected answer/result: Criteria/formula: Rows=ClassName; Columns=Gender; Value=Count(StudentID)
Result: Counts students by class and gender.

E16. Q_LargeLowCost — 1 mark

1. Add ACTIVITY.
2. Add ActivityName, Participants and Fee.
3. On one row use >=50 and <=1000.

4. Run and save as Q_LargeLowCost.

   Expected answer/result: Criteria/formula: Participants>=50 AND Fee<=1000
   Result: At least 50 participants and fee not above 1000.

E17. Q_ICT_Academic — 2 marks

1. Add ACTIVITY.
2. Add the needed fields.
3. For ActivityDate use Between #01/02/2026# And #31/03/2026#.
4. Use Category "ICT" on one row and "Academic" on the next row.
5. Run and save as Q_ICT_Academic.

   Expected answer/result: Criteria/formula: Date range AND (Category="ICT" OR Category="Academic")
   Result: Correct date range plus ICT or Academic.

E18. Q_ActivityPattern — 1 mark

1. Add ACTIVITY.
2. Add ActivityName.
3. Use Like "C*" on one row.
4. Use Like "*Day" on the OR row.
5. Run and save as Q_ActivityPattern.

   Expected answer/result: Criteria/formula: Like "C*" OR Like "*Day"
   Result: Starts with C OR ends with Day.

E19. Q_TotalCollection — 2 marks

1. Add ACTIVITY.
2. Add ActivityName, Participants and Fee.
3. Create TotalCollection.
4. Use [Participants]*[Fee].
5. Set Criteria to >50000.
6. Run and save as Q_TotalCollection.

   Expected answer/result: Criteria/formula: TotalCollection=[Participants]*[Fee]; >50000
   Result: Participants × Fee is greater than 50,000.

E20. Q_ParticipantsORFree — 1 mark

1. Add ACTIVITY.
2. Add the needed fields.
3. Use Between 30 And 80 for Participants.
4. On the OR row use 0 for Fee.
5. Run and save as Q_ParticipantsORFree.

   Expected answer/result: Criteria/formula: Participants Between 30 And 80 OR Fee=0
   Result: 30–80 participants OR free activity.

Query marking: for 2 marks, normally 1 mark for correct fields/design and 1 mark for correct criteria/result. For 3 marks, give the extra mark for the calculated field/full correct design.

QUESTION F — SORTING AND FILTERING (5 MARKS)

F(a) Q_TopFive — 2 marks

1. Create a query from STUDENT.
2. Add the needed fields and Mark.
3. Set Mark Sort to Descending.
4. Set Top Values = 5.
5. Run and save as Q_TopFive.

   Expected answer/result: Top five marks: 91, 88, 84, 78, 76.

F(b) Q_FormII — 1 mark

1. Create a query from STUDENT.
2. Add the needed fields.
3. Under ClassCode type In ("F2A","F2B").
4. Run and save as Q_FormII.

   Expected answer/result: Only F2A and F2B students.

F(c) Q_ActivityDate — 1 mark

1. Create a query from ACTIVITY.
2. Add ActivityName and ActivityDate.
3. Set ActivityDate Sort to Ascending.
4. Run and save as Q_ActivityDate.

   Expected answer/result: Earliest date first.

F(d) Q_MostParticipants — 1 mark

1. Create a query from ACTIVITY.
2. Add ActivityName and Participants.
3. Set Participants Sort to Descending.
4. Run and save as Q_MostParticipants.

   Expected answer/result: Highest participant number first.

QUESTION G — FORMS (6 MARKS)

G(a) FORM_STUDENT — Columnar — 2 marks

1. Select the STUDENT table.
2. Go to Create → Form Wizard.
3. Select the required fields.
4. Choose Columnar layout.
5. Name the form FORM_STUDENT.
6. Finish and save.

   Expected answer/result: A working form named FORM_STUDENT using Columnar layout.

G(b) FORM_CLASS — Tabular — 2 marks

1. Select the CLASS table.
2. Go to Create → Form Wizard.

3. Select the required fields.
4. Choose Tabular layout.
5. Name the form FORM_CLASS.
6. Finish and save.

   Expected answer/result: A working form named FORM_CLASS using Tabular layout.

G(c) FORM_ACTIVITY — Justified — 2 marks

1. Select the ACTIVITY table.
2. Go to Create → Form Wizard.
3. Select the required fields.
4. Choose Justified layout.
5. Name the form FORM_ACTIVITY.
6. Finish and save.

   Expected answer/result: A working form named FORM_ACTIVITY using Justified layout.

QUESTION H — REPORTS (6 MARKS)

H(a) REPORT_STUDENT_CLASS — 2 marks

1. Select Q_FormIII.
2. Open Report Wizard.
3. Add FirstName, LastName, ClassName, Stream and Mark.
4. Group by ClassName.
5. Sort Mark Highest to Lowest.
6. Save as REPORT_STUDENT_CLASS.

   Expected answer/result: Form III students grouped by ClassName, with marks highest first.

H(b) REPORT_ACTIVITY_CATEGORY — 2 marks

1. Select ACTIVITY.
2. Open Report Wizard.
3. Select ActivityName, Participants and Fee.
4. Group by Category.
5. Save as REPORT_ACTIVITY_CATEGORY.

   Expected answer/result: Activities grouped by category.

H(c) REPORT_PERFORMANCE — 2 marks

1. Select Q_Performance.
2. Open Report Wizard.
3. Select student name, Mark and PerformanceBand.
4. Give the report a clear title.
5. Add a page number in the footer.
6. Save as REPORT_PERFORMANCE.

   Expected answer/result: Shows student name, Mark and PerformanceBand, with title and page number.

QUESTION I — TESTING (6 MARKS)

No.	Value to test	Expected result	Reason
1	Mark = 105	Reject	Mark must be 0–100.
2	Participants = 0	Reject	Participants must be at least 1.
3	Fee = -500	Reject	Fee cannot be negative.
4	Gender = Other	Reject	Only Male or Female is allowed.
5	Phone = 07123456	Reject	Phone must have 10 digits and start with 0.
6	ClassCode = F9A	Reject	F9A is not in CLASS; relationship is enforced.

1. Open the correct table in Datasheet View.
2. Enter the wrong value.
3. Check that Access rejects it.
4. Cancel the wrong entry and return a valid value.

Expected answer/result: All six tests must give Reject.

QUESTION J — DATABASE CHECK (2 MARKS)

J(a) Check saved objects — 1 mark

1. Look at the Navigation Pane.
2. Check STUDENT, CLASS and ACTIVITY.
3. Check the one relationship.
4. Check all 20 queries.
5. Check 3 forms and 3 reports.

J(b) Check names — 1 mark

Expected answer/result: All object names must be exactly as required in the examination.

Object	Required names
Tables	STUDENT, CLASS, ACTIVITY
Forms	FORM_STUDENT, FORM_CLASS, FORM_ACTIVITY
Reports	REPORT_STUDENT_CLASS, REPORT_ACTIVITY_CATEGORY, REPORT_PERFORMANCE

The 20 query names are listed on the Quick Answer Sheet below.

QUESTION K — FINAL SUBMISSION (2 MARKS)

K(a) Preview reports — 1 mark

1. Open each report in Print Preview.
2. Check headings and data.
3. Make sure no column or text is cut off.

K(b) Save and submit — 1 mark

1. Save all objects.
2. Close the database if required.
3. Submit the database file as instructed by the teacher.

   Expected answer/result: Database is complete, saved and ready for marking.

QUICK ANSWER SHEET — ALL QUERY CRITERIA

Query	Criteria / formula
Q_AllStudents	No criteria
Q_NameStartsM	Like "M*"
Q_LastNameAR	Like "*ar*"
Q_ClassIN	In ("F2A","F2B","F3A")
Q_Mark75Plus	>=75
Q_Mark60_80	Between 60 And 80
Q_Female70Plus	Gender="Female" AND Mark>=70
Q_MaleOR90	Gender="Male" OR Mark>=90
Q_DOB_2009_2010	Between #01/01/2009# And #31/12/2010#
Q_Older70Plus	DOB < #01/01/2010# AND Mark>=70
Q_Age16Plus	DateDiff("yyyy",[DOB],Date()) - IIf(Format([DOB],"mmdd")>Format(Date(),"mmdd"),1,0) ; >=16
Q_FormIII	ClassName="Form III"
Q_Performance	IIf([Mark]>=80,"Excellent",IIf([Mark]>=60,"Good","Needs Improvement"))
Q_MinimumMark	[Enter minimum Mark:]
Q_ClassGender	Crosstab: ClassName rows; Gender columns; Count(StudentID)
Q_LargeLowCost	Participants>=50 AND Fee<=1000
Q_ICT_Academic	Date range AND (Category="ICT" OR Category="Academic")
Q_ActivityPattern	Like "C*" OR Like "*Day"
Q_TotalCollection	[Participants]*[Fee] ; >50000
Q_ParticipantsORFree	Participants Between 30 And 80 OR Fee=0

FINAL MARK SUMMARY

Section	Marks
A	5
B	20
C	8
D	10
E	30
F	5
G	6
H	6

Section	Marks
I	6
J	2
K	2
TOTAL	100

END OF MARKING SCHEME